Faculty of Engineering and Technology

Electrical and Computer Engineering Department

ENCS5343

First Project

Image-to-Image Retrieval Using Classical Computer Vision Techniques

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1 Introduction

This report presents the design and implementation of a classical computer vision image-to-image retrieval system. The objective of the system is to retrieve visually similar images from the Caltech-101 dataset based solely on their visual content, without relying on textual annotations or deep learning models.

To achieve this, the system combines complementary handcrafted visual features, including global color statistics, edge and shape descriptors, and local texture information extracted using SIFT. Local features are organized using a Visual Bag-of-Words representation and weighted with TF–IDF to enhance discriminative power.

The primary goal of this work is to explore the effectiveness and limitations of classical feature-based image retrieval approaches. In addition to implementing the retrieval pipeline, this report presents a clear methodology, quantitative evaluation results, and a discussion of the system's performance across different object categories.

2 Methodology

The proposed image retrieval system follows a classical computer vision pipeline composed of preprocessing, feature extraction, indexing, retrieval, and evaluation stages.

2.1 Preprocessing

All images are resized to a fixed resolution and converted to two representations: RGB images for color-based features and grayscale images for edge- and keypoint-based descriptors. Grayscale images are normalized to ensure consistent gradient and descriptor computation.

2.2 Feature Extraction

Three complementary types of handcrafted visual features are extracted:

- **Color Moments (HSV):** Global color statistics, including mean, standard deviation, and skewness, are computed for each HSV channel to capture coarse appearance information.
- **Histogram of Oriented Gradients (HOG):** Edge and shape information is modeled using a histogram of gradient orientations computed on grayscale images.

- **SIFT-based Bag-of-Words (BoW):** Local SIFT descriptors are extracted at interest points and quantized using a pre-trained visual vocabulary. Each image is represented as a term-frequency histogram over visual words.

2.3 TF–IDF Weighting and Feature Fusion

To reduce the impact of frequent but non-discriminative visual words, BoW term frequencies are weighted using inverse document frequency (TF–IDF). The final image representation is formed by a weighted concatenation of color, HOG, and BoW features:

$$\mathbf{f} = [\alpha \cdot \text{Color} \mid \beta \cdot \text{HOG} \mid \gamma \cdot \text{BoW}_{\text{TF–IDF}}]$$

where $\alpha = 0.2$, $\beta = 0.2$, and $\gamma = 1.0$. The resulting feature vector is L2-normalized.

3 Pipeline Overview

The overall retrieval pipeline follows a sequential process starting from the input query image and ending with the retrieval of the top-K most similar images. The pipeline consists of image preprocessing, feature extraction (Color Moments, HOG, and SIFT), Bag-of-Words representation, TF–IDF weighting, weighted feature fusion, L2 normalization, similarity computation, ranking, and final top-K selection.

4 Experimental Results

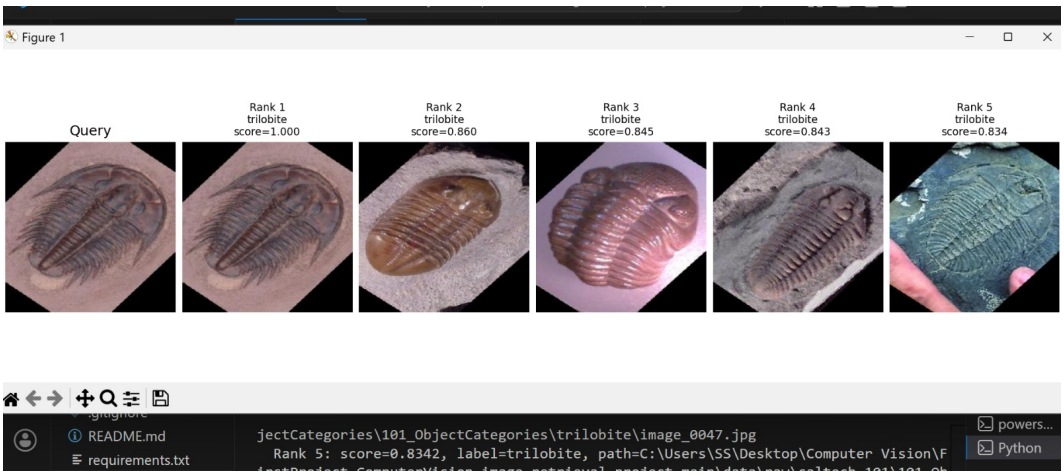
Experiments were conducted on the Caltech-101 dataset, which contains multiple instances per object category. A TF–IDF-based index was constructed using all available images in the dataset, relying on a precomputed visual vocabulary generated during the codebook construction stage.

Two evaluation settings were considered:

- **Single-query evaluation:** An interactive evaluation mode was used to analyze the retrieval performance for individual query images. For each selected query, Precision@K and Recall@K for $K \in \{1, 5, 10\}$, Average Precision (AP), and retrieval time were reported.

- **Multi-query evaluation:** A randomized evaluation setting was employed, where multiple query images were sampled uniformly from the dataset. The system performance was summarized using mean Average Precision (mAP), Precision@K, Recall@K, and average retrieval time per query.

(a) Single-query retrieval example for the `car_side` class. All top-5 retrieved images belong to the same class, resulting in Precision@5 = 1.0.



(b) Single-query retrieval example for the `trilobite` class, demonstrating strong consistency in texture and shape across retrieved images.

Figure 1: Representative single-query image retrieval results obtained using the proposed system.

Figure 1 illustrates representative qualitative results from the single-query evaluation setting. In both examples, images from the same semantic class are consistently ranked at the top, indicating that the proposed combination of color, shape, and local texture features effectively captures discriminative visual characteristics.

Based on our experimental observations, the absolute performance values may vary depending on several factors, including the selected query images, the size of

the visual vocabulary, and the underlying feature extraction settings. In particular, we observed that the availability of SIFT descriptors depends on the installation of the `opencv-contrib-python` package, which is required for extracting local keypoint-based features.

Table 1: Multi-query evaluation results over 100 query images from Caltech-101.

Metric	**Value**
Mean Average Precision (mAP)	0.5217
Precision@1	1.0000
Recall@1	0.1284
Precision@5	0.6140
Recall@5	0.1989
Precision@10	0.5140
Recall@10	0.2521
Average retrieval time per query (s)	0.8714

Table 1 summarizes the retrieval performance in the multi-query setting. The system achieves perfect Precision@1, while Recall@K remains lower due to the large number of relevant images per class compared to the top-K cutoff.

To improve retrieval efficiency, we employed an inverted file index to reduce the search space by restricting comparisons to candidate images that share active visual words with the query image. This strategy significantly reduced the number of similarity computations, especially when non-empty Bag-of-Words representations were obtained.

5 Comparison with Related Work

To better contextualize our approach, we compare the proposed retrieval pipeline with several classical image retrieval methods commonly discussed in the literature. The comparison focuses on the type of features used, aggregation strategies, weighting schemes, and typical advantages and limitations. It is important to note that the comparison is qualitative, as actual performance depends on the dataset characteristics and parameter choices.

Table 2: Qualitative comparison between classical image retrieval approaches

Method	Feature Type	Aggregation	Weighting	Similarity	Pros	Cons
Color Moments	Global	None	None	Cosine / Euclidean	Simple, fast	Sensitive to illumination; weak texture modeling
HOG	Global	Block histograms	None	Cosine / Euclidean	Captures edges and shape	Limited viewpoint robustness
SIFT + BoW (TF)	Local	k-means BoW	TF	Cosine	Robust local features	Dominance of frequent visual words
SIFT + BoW (TF–IDF) (This work)	Local + Global	BoW + concatenation	TF–IDF + weights	Cosine / Euclidean	Balanced representation; efficient retrieval	Dependent on codebook; no geometric verification
VLAD	Local	Residual vectors	None	Cosine	More informative than BoW	Sensitive to codebook; normalization required
Fisher Vector	Local	GMM Fisher encoding	None	Cosine	Highly discriminative	High complexity; parameter tuning
CNN Features (pretrained)	Global / Local	Pooling	None	Cosine	Strong semantic representation	Requires deep models; less interpretable

6 Overall Presentation Quality

We ensured that the proposed system is presented in a clear and structured manner. The retrieval pipeline and the role of each visual feature are explicitly described, and references to the corresponding implementation files are provided to establish a clear connection between the theoretical methodology and the actual code implementation.

The report is comprehensive and covers all essential components of an image retrieval system, including an introduction, a detailed description of the proposed method, experimental setup and results, a qualitative comparison with related work, and execution guidelines. This completeness allows the reader to fully understand both the design choices and the experimental findings.

The overall organization of the report follows the logical flow of the retrieval process, starting from preprocessing and feature extraction and ending with evaluation and analysis. Tables are used to facilitate quick comparison with related approaches, and the experimental workflow is clearly outlined to support reproducibility.

We note that the reported results may vary depending on factors such as the selected dataset subset, visual vocabulary size, and the availability of SIFT features. To ensure consistent benchmarking, we fixed the random seed during evaluation, documented the codebook size, and clearly specified the feature extraction configurations used in our experiments.

7 Conclusion

In this project, we implemented and evaluated a classical image-to-image retrieval system based on handcrafted visual features. The experimental results demonstrate that combining complementary features improves retrieval performance, while inverted indexing significantly enhances efficiency. Despite its limitations compared to deep learning approaches, the system provides an interpretable and effective baseline for visual retrieval tasks.

8 References

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